Qns	Answer	Marks
5(a)	Rate of change of angular displacement	B1
5(b)(i)	At minimum speed, normal contact force = 0 AND Weight provides centripetal force	B1
	$mg = \frac{mv^2}{r}$ $9.81 = \frac{v^2}{0.12}$	C1
	$v = 1.08$ or $1.1\ \text{m s}^{-1}$	A1
5(b)(ii)	Loss in EPE = Gain in GPE + Gain in KE	B1
	$\frac{1}{2}kx^2 = mgh + \frac{1}{2}mv^2$	
	$\frac{1}{2}k(0.16)^2 = (0.076)(9.81)(0.24) + \frac{1}{2}(0.076)(1.1)^2$	C1
	$k = 17.6 = 18\ \text{N m}^{-1}$	A1
	OR	
	$\frac{1}{2}k(0.16)^2 = (0.076)(9.81)(0.24) + \frac{1}{2}(0.076)(1.08)^2$	(C1)
	$k = 17.4 = 17\ \text{N m}^{-1}$	(A1)
5(c)	Loss in KE = Work done against resistive force	B1
	0.23 = (average resistive force)$\left(0.30 + 0.25 + 2\pi(0.12)\right)$	
	average resistive force = 0.18 N	A1